

Ion channel gating

Name _____ Date _____

Use this sheet while you work through the ion channel gating walkthrough. It has 13 predictions.

1. When a step asks you to predict, write or draw your prediction in the top box, in pencil or your first color. Type it into the worksheet beside the walkthrough too. Next stays locked until you press Show me.
2. Press Show me. In the bottom box, in a second color, write what actually happened and whether you were right. If you were not, write what you would change.
3. When you finish the walkthrough, close it and do Part 2 from memory. Then reopen it and correct Part 2 in your second color. What you had to fix is what to study.

Part 1. Your predictions

PREDICTION 1 OF 13 · MOST DOORS HAVE A GATE

A cell has to fire a signal at one moment and stay quiet the next. Why would a gate that opens only when told be more useful than a door that is always open?

My prediction, before Show me

What happened, and was I right?

PREDICTION 2 OF 13 · THE ONE THAT ANSWERS TO VOLTAGE

Voltage-gated channels open when the membrane becomes less negative. Once they open and let positive ions in, the membrane becomes less negative still. What does that do to the neighboring voltage-gated channels?

My prediction, before Show me

What happened, and was I right?

PREDICTION 3 OF 13 · THE ONE THAT WAITS FOR A CHEMICAL

At the junction between a motor neuron and a muscle fiber, the neuron releases acetylcholine and the muscle's channels open. What would happen at that junction if a drug sat in those binding sites without opening them?

My prediction, before Show me

What happened, and was I right?

PREDICTION 4 OF 13 · THE ONE THAT ANSWERS TO TOUCH

These channels sit in the sensory endings in your skin, in the hair cells of your inner ear, and in the walls of your blood vessels. What do those three places have in common?

My prediction, before Show me

What happened, and was I right?

PREDICTION 5 OF 13 · NOT ONE GATE, TWO

Two gates on one pore, and both must be open for anything to pass. How many different states can that channel be in?

My prediction, before Show me

What happened, and was I right?

PREDICTION 6 OF 13 · THRESHOLD, AND IT OPENS

Sodium is far more concentrated outside the cell, and the inside is negative. Both of those push sodium the same direction. What happens to the voltage once the pore is clear?

My prediction, before Show me

What happened, and was I right?

PREDICTION 7 OF 13 · THE SECOND GATE SWINGS SHUT

The outer gate is still open at this moment, but the pore is blocked from the inside. Can sodium get through, and can the cell open this channel again right now?

My prediction, before Show me

What happened, and was I right?

PREDICTION 8 OF 13 · WHY THAT MATTERS SO MUCH

For a short period after firing, most sodium channels in that patch of membrane are inactivated and cannot reopen. What does that do to a second signal arriving right behind the first one?

My prediction, before Show me

What happened, and was I right?

PREDICTION 9 OF 13 · RESETTING THE CHANNEL

A channel resets only when the membrane repolarizes. What would happen to a neuron sitting in a fluid that kept it permanently depolarized, never letting it return toward its resting voltage?

My prediction, before Show me

What happened, and was I right?

PREDICTION 10 OF 13 · LATE TO OPEN, LATE TO CLOSE

Potassium channels open late and stay open a little too long, letting potassium keep leaving after the voltage has already come back down to rest. What does the voltage do in that moment?

My prediction, before Show me

What happened, and was I right?

PREDICTION 11 OF 13 · PATIENT: THE DENTIST'S INJECTION

Lidocaine blocks the sodium channel pore. Using the three states you just built, explain why the patient stops feeling pain in that area.

My prediction, before Show me

What happened, and was I right?

PREDICTION 12 OF 13 · PATIENT: WEAKNESS AFTER A BIG MEAL

His sodium channels are slow to inactivate, which leaves the muscle membrane depolarized for longer than it should be. Why would that cause weakness rather than extra contraction?

My prediction, before Show me

What happened, and was I right?

PREDICTION 13 OF 13 · THE WHOLE STORY, WITH THE SCIENCE NAMES

Without scrolling back, name the three states of a voltage-gated sodium channel, say what each gate is doing in each state, and say what has to happen before the channel can fire again.

My prediction, before Show me

What happened, and was I right?

Part 2. Draw it from memory

Channel gating

1. Four channels, four behaviors

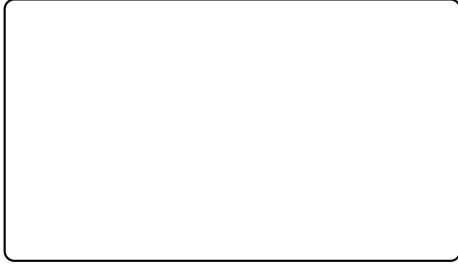
Draw a leak channel, a voltage-gated channel, a chemically gated channel, and a mechanically gated channel. Under each, write what opens it.

Leak	Voltage gated
Chemically gated	Mechanically gated

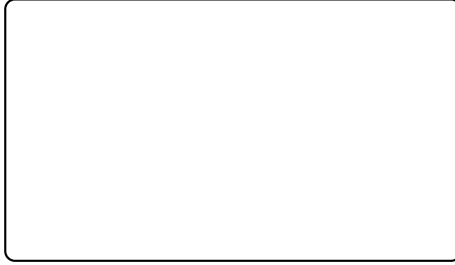
2. The sodium channel in all three states

Draw the same voltage-gated sodium channel three times. Show both gates in each drawing. Under each one, write the name of the state, the membrane voltage at that moment, and whether sodium is moving.

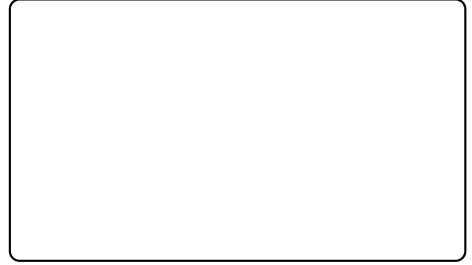
State 1



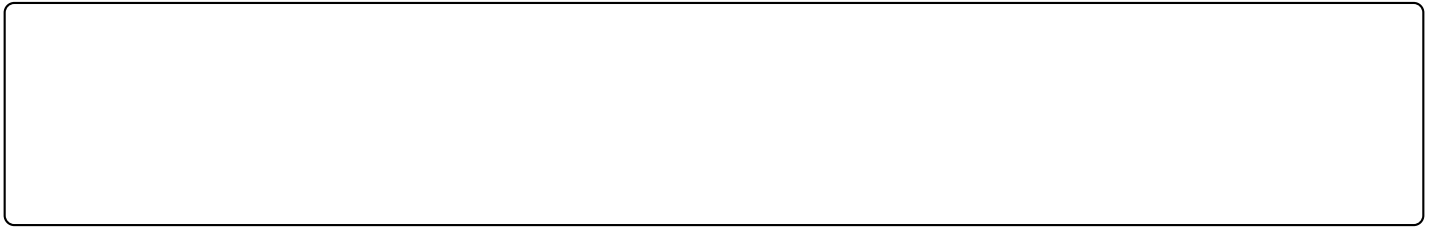
State 2



State 3



Then answer: what has to happen before the channel can fire again, and why can a stronger stimulus not force it?



3. Read a patient

A patient's blood potassium is 7.0 mmol/L (normal range 3.5 to 5.0 mmol/L). Explain, in the order the events happen, why this makes excitable tissue stop responding rather than fire more. Use the three channel states in your answer.

